

B.TECH ELECTRICAL ENGINEERING

Summer Examination- ~~April 2016~~ Nov 2016

Subject: High Voltage Engineering

Semester: VIII

Date: ~~17 May 2016~~ 07/11/2016

Time: 3Hours

Marks: 70

INSTRUCTIONS:

- 1] Question one is compulsory.
- 2] Attempt any **Five** questions from the remaining.
- 3] Assume suitable data wherever necessary and mention it clearly. If it is not given, then it is part of examination.

- Q. No. 1 Fill in the blanks with the correct answer. [10]**
- (a) Loss of charge method is used to determine _____.
 - (b) The electrical field developed within clouds before a lightning stroke occurs can be of the order of _____.
 - (c) Surge impedance ^{of} loss less transmission line is _____.
 - (d) The equivalent circuit of a surge arrester may be represented as _____.
 - (e) The material used in gapless surge arrester for HV system is _____.
- Q. No. 2 Answer the following questions. [Any Two] [6]**
- (a) What is "stressed oil volume theory", how does it explain breakdown in large volumes of commercial liquid dielectrics? [6]
 - (b) In an experiment in a certain gas it was found that the steady state current is $5.5 \times 10^{-8} \text{A}$ at 8kV at a distance of 0.4cm between the plane electrodes. Keeping the field constant and reducing the distance to 0.1cm results in a constant of $5.5 \times 10^{-9} \text{A}$. Calculate Townsend's primary ionizations coefficient α . [6]
 - (c) Discuss the various mechanisms of vacuum breakdown. [6]
- Q. No. 3 Attempt the following questions.[Any Two] [6]**
- (a) Explain the different mechanisms by which breakdown occurs in solid dielectrics? [6]
 - (b) What is a composite dielectric and what are its properties? [6]
 - (c) What are insulation requirements for high-voltage Air or SF₆ circuit breakers? [6]
- Q. No. 4 Answer the following questions. [Any Two] [6]**
- (a) Describe, with a neat sketch, the working of Van de Graaff generator, what are the factors that limit the maximum voltage obtained? [6]
 - (b) Why is it preferable to use isolating transformers for excitation with cascade transformer units, if the power requirement is large? [6]
 - (c) What is a Tesla coil? How is damped high-frequency oscillations obtained from a Tesla coil? [6]
- Q. No.5 Solve the following questions.[Any Two] [6]**
- (a) Explain the importance of switching over voltages in EHV power system. How is protection against over voltages achieved? [6]

- (b) Discuss the different methods of measuring high dc voltages. What are the limitations in each method? [6]
- (c) Design a peak reading voltmeter along with a suitable microammeter such that it will be to read voltages, up to 100kV (peak). The capacitance potential divider available is of the ratio 1000:1. [6]

Q. No. 6 Attempt the following questions.

- (a) A 3- Φ single circuit transmission line is 400km long. If the line is rated for 220kV and has the parameter, $R=0.1\Omega/\text{km}$, $L=1.26\text{mH}/\text{km}$, $C=0.009\mu\text{F}/\text{km}$, and $G=0$, find (i) the surge impedance, and (ii) the velocity of propagation neglecting the resistance of the line. If a surge of 150kV and infinitely long tail strikes at one end of the line, what is the time taken for the surge to travel to the other end of the line? [6]
- (b) What are the causes for switching and power frequency over voltages? How are they controlled in power system? [6]

Q. No. 7 Answer the following questions.

- (a) Explain the terms (i) disruptive discharge voltage, (ii) Impulse voltages [6]
- (b) Discuss the testing of bushing conducted as per the IS standard. [6]

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